

Support Ticket

#736196 - Create a custom Docker image

Posted on Thursday 10th April at 04:44

Objective:

This task involves creating a Docker image for a web application, deploying the image as a container, and configuring the container to use a specific working directory and port.

Scenario:

Your company is adopting a container-based approach for deploying applications. This task focuses on creating a Docker image for a web application and deploying it as a container with specific configurations.

Note: Use the same website which you hosted in the previous task.

Completion Criteria:

- Use docker.io/redhat/ubi9 as a base image
- Docker image is created with the web application.
- Make sure a Docker image has a 18085/tcp as an exposed port.
- Make sure a Docker image has a /tmp/appdata as a Working Directory.

Task Summary

- **Base Image:** docker.io/redhat/ubi9
- **Exposed Port:** 18085/tcp
- **Working Directory:** /tmp/appdata
- **Web App:** Use the same one from your previous task (e.g., the one hosted under mytask.intern.com)

Create a directory structure

```
controlplane:~$ mkdir /webapp
controlplane:~$ cd /webapp/
controlplane:/webapp$ vi Dockerfile
controlplane:/webapp$ ls
Dockerfile
```

Dockerfile

```
FROM docker.io/redhat/ubi9

RUN yum update -y && yum install httpd -y && yum install wget -y && yum install zip -y
RUN wget https://www.free-css.com/assets/files/free-css-templates/download/page296/carvilla.zip
RUN unzip carvilla.zip -d /mnt
RUN cp -rvf /mnt/carvilla-v1.0/* /var/www/html

EXPOSE 80

CMD ["httpd","-D","FOREGROUND"]
```

Build the Docker Image

Navigate to your project directory and run:

\$ docker build -t myimg .

```
controlplane:/webapp$ docker build -t myimg .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
             Install the buildx component to build images with BuildKit:
             https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  2.048kB
Step 1/7 : FROM docker.io/redhat/ubi9
latest: Pulling from redhat/ubi9
e98f4315d8c9: Pull complete
15b7d555f3bb: Pull complete
Digest: sha256:ea57285741f007e83f2ee20423c20b0cbcce0b59cc3da027c671692cc7efe4dd
Status: Downloaded newer image for redhat/ubi9:latest
--> d53e845ae020
Step 2/7 : RUN yum update -y && yum install httpd -y && yum install wget -y && yum install zip -y
--> Running in 5ae104eaa6af
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use subscription-manager to register.

Red Hat Universal Base Image 9 (RPMs) - BaseOS 353 kB/s | 526 kB 00:01
Red Hat Universal Base Image 9 (RPMs) - AppStre 1.4 MB/s | 2.3 MB 00:01
```

Check Image created

```
controlplane:/webapp$ docker images
REPOSITORY      TAG          IMAGE ID      CREATED        SIZE
myimg           latest      532ee4a06333  About a minute ago  247MB
redhat/ubi9     latest      d53e845ae020  5 days ago    205MB
controlplane:/webapp$
```

Run the Docker Container

Now run the container with port mapping:

\$ docker run --name app1 -p 18085:80 -itd myimg

```
controlplane:/webapp$ docker run --name app1 -p 18085:80 -itd myimg
c284096463eb8c8ab9c9d07861ae2f309bad8d318f54803dc6463fe2289067fb
controlplane:/webapp$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
c284096463eb   myimg     "httpd -D FOREGROUND"   9 seconds ago Up 8 seconds   0.0.0.0:18085->80/tcp, :::18085->80/tcp   app1
controlplane:/webapp$
```

Access the PORT 18085.

